



what percent of manure is carbon dry weight?

Manure typically contains 30-45% carbon by dry weight, varying by animal type and handling. This range reflects organic carbon in livestock feces, with chicken manure around 33%, dairy at 38%, swine at 41%, and cattle at 43%.^[1]

Variation by Animal Type

Carbon content differs across species due to diet, digestion, and bedding.

Animal Type	% Carbon (dry weight)
Chicken	32.5% ^[1]
Dairy	38.2% ^[1]
Swine	40.6% ^[1]
Cattle	43.0% ^[1]

Solid dairy manure samples average 35.6% total carbon on a dry basis, ranging 18-44%.^[2]

Estimation Methods

Carbon is often half the volatile solids (VS) content, as organic matter in manure is roughly 50% carbon. Models assume ~42% for nutrient calculations. For soil carbon modeling like RothC, use type-specific values from lab analysis.^{[3] [4] [5] [2]}



Is 50% a good assumption for carbon percentage?

No, 50% is somewhat high as a general assumption for the carbon percentage in manure dry weight. Typical values range from 30-45%, so 50% may overestimate carbon inputs in models like RothC.^{[19] [20]}

Why It's High

Literature consistently reports lower averages, such as 38% for dairy and up to 43% for cattle manure on a dry basis. A 50% figure might stem from approximating organic matter as 50% carbon, but manure's organic fraction isn't 100% of dry matter (includes ash/minerals at 10-20%).^{[20] [21] [19]}

When to Use Alternatives

- For conservative estimates in carbon accounting (e.g., Verra VM0042), stick to 35-40% or site-specific data.^[20]
- If assuming VS $\approx$ 75-85% of dry matter and VS $\approx$ 50% C, effective %C $\approx$ 37.5-42.5%.^[20] Use lab analysis for precision in tea/grassland projects.^[22]



So let's say I'm assuming an application of 0.5 tons/ha of tons of Carbon per hectare ... how much manure would that entail, if we say 40% of manure is carbon?

To apply 0.5 tons (500 kg) of carbon per hectare using manure at 40% carbon content, you'd need 1.25 tons (1250 kg) of dry manure per hectare.

Calculation

Carbon required = 0.5 tons/ha.

Manure dry weight needed = carbon required $\div$ carbon fraction = 0.5 tons $\div$ 0.4 = 1.25 tons/ha.^[23]
^[24]

This assumes dry weight basis; fresh manure requires adjustment for moisture (typically 70-85% water), scaling up to 8-25 tons/ha wet weight depending on solids content.^[24]

Context for Modeling

In soil carbon models like RothC or Verra VM0042 projects, apply this rate uniformly or via partitioning (e.g., 50% to SOC pool). Verify with lab analysis for your tea/grassland sites, as 40% aligns well with dairy/cattle averages.^[23]



Is that a reasonable amount of manure for Kenyan farmlands?

No, 1.25 tons/ha of dry manure is unrealistically low and not reasonable for Kenyan farmlands. Kenyan smallholders typically apply 5-10 tons/ha (wet weight) of cattle manure for field crops like maize or tea, or 20-40 tons/ha for high-value crops.^[35] ^[36] ^[37]

Why It's Low

Your 1.25 tons/ha dry equates to just 6-12 tons/ha wet (at 80-90% moisture), but studies show actual rates are higher to meet nutrient needs on nutrient-poor soils. Machakos trials found no maize yield gain above 40 tons/ha wet boma manure, with economics favoring 20 tons/ha equivalents.^{[36] [35]}

Availability Context

Central Kenya highlands generate 5-10 tons/ha potential from on-farm livestock (0.5-1 cow equivalents), but collection/composting losses limit usable amounts to 2-5 tons/ha annually. Importing more is common but costly for carbon projects. For RothC modeling in tea/grasslands, calibrate to local wet rates and measure SOC response.^{[38] [39] [40]}



Is that a reasonable amount in terms of dry weight?

Yes, 1.25 tons/ha dry manure is reasonable in terms of dry weight for Kenyan farmlands. It aligns with conservative or targeted applications in smallholder systems, especially for carbon-focused projects rather than high nutrient demands.^{[45] [46] [47]}

Dry Weight Benchmarks

Kenyan studies report dry manure equivalents of 1-2 tons/ha from typical wet applications (5-10 t/ha at 20% dry matter). One trial used 8 Mg/ha (8 t/ha) dry cattle manure for maize, but averages are lower at ~2 t/ha dry from on-farm sources.^{[46] [48] [45]}

Practicality for Your Context

For tea/grasslands in carbon modeling (RothC/Verra VM0042), 1.25 t/ha dry delivers precise C inputs without excess nutrients, feasible via composting zero-grazing dairy manure (20% DM). Higher rates (e.g., 4 t/ha dry) risk P buildup on Kenyan soils.^{[49] [45]}



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